



Advanced Linear Algebra: Foundations to Frontiers

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7.5.2 Summary ¶

Let $A \in \mathbb{R}^{n \times n}$ be tridiagonal and SPD so that

$$A = \begin{pmatrix} \alpha_{0,0} & \alpha_{1,0} & & & \\ \alpha_{1,0} & \alpha_{1,1} & \alpha_{2,1} & & \\ & \ddots & \ddots & \ddots & \\ & & \alpha_{n-2,n-3} & \alpha_{n-2,n-2} & \alpha_{n-1,n-2} \\ & & & \alpha_{n-1,n-2} & \alpha_{n-1,n-1} \end{pmatrix}.$$

Then its Cholesky factor is given by

$$\begin{pmatrix} \lambda_{0,0} & & & & \\ \lambda_{1,0} & \lambda_{1,1} & & & \\ & \ddots & \ddots & \ddots & \\ & & \lambda_{n-2,n-3} & \lambda_{n-2,n-2} & \\ & & & \lambda_{n-1,n-2} & \lambda_{n-1,n-1} \end{pmatrix}.$$

An algorithm for computing it is given by

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for  $i = 0, \dots, n-2$ 
   $\alpha_{i,i} := \sqrt{\alpha_{i,i}}$ 
   $\alpha_{i+1,i} := \alpha_{i+1,i} / \alpha_{i,i}$ 
   $\alpha_{i+1,i+1} := \alpha_{i+1,i+1} - \alpha_{i+1,i} \alpha_{i+1,i}$ 
endfor
 $\alpha_{n-1,n-1} := \sqrt{\alpha_{n-1,n-1}}$ 

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It requires n square roots, $n-1$ divides, $n-1$ multiplies, and $n-1$ subtracts. An algorithm for overwriting y with the solution to $Ax = y$ given its Cholesky factor is given by

- Overwrite y with the solution of $Lz = y$ (forward substitution) is accomplished by the following algorithm (here L has overwritten A):

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for  $i = 0, \dots, n-2$ 
   $\psi_i := \psi_i / \alpha_{i,i}$ 
   $\psi_{i+1} := \psi_{i+1} - \alpha_{i+1,i} \psi_i$ 
endfor
 $\psi_{n-1} := \psi_{n-1} / \alpha_{n-1,n-1}$ 

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- Overwriting y with the solution of $L^T x = z$, (where z has overwritten y (back substitution).

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for  $i = n-1, \dots, 1$ 
   $\psi_i := \psi_i / \alpha_{i,i}$ 
   $\psi_{i-1} := \psi_{i-1} - \alpha_{i,i-1} \psi_i$ 
endfor
 $\psi_0 := \psi_0 / \alpha_{0,0}$ 

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Definition 7.5.2.1. The half-band width of a symmetric matrix equals the number of subdiagonals beyond which all the matrix contains only zeroes. For example, a diagonal matrix has half-band width of zero and a tridiagonal matrix has a half-band width of one.

Nested dissection: a hierarchical partitioning of the graph that captures the sparsity of a matrix in an effort to reorder the rows and columns of that matrix so as to reduce fill-in (the overwriting of zeroes in the matrix with nonzeros).

Splitting methods: The system of linear equations $Ax = b$, splitting methods view A as $A = M - N$ and then, given an initial approximation $x^{(0)}$, create a sequence of approximations, $x^{(k)}$ that under mild conditions converge to x by solving

$$Mx^{(k+1)} = Nx^{(k)} + b$$

or, equivalently, computing

$$x^{(k+1)} = M^{-1}(Nx^{(k)} + b).$$

This method converges to x if for some norm $\|\cdot\|$

$$\|M^{-1}N\| < 1.$$

Given $A = D - L - U$ where $-L$, D , and $-U$ equal the strictly lower triangular, diagonal, and strictly upper triangular parts of A , commonly used splitting methods are

- Jacobi iteration: $A = \underbrace{D}_M - \underbrace{(L+U)}_N$.
- Gauss-Seidel iteration: $A = \underbrace{D-L}_M - \underbrace{U}_N$.
- Successive Over-Relaxation (SOR): $A = \underbrace{\frac{1}{\omega}D-L}_M - \underbrace{\left(\frac{1-\omega}{\omega}D+U\right)}_N$, where ω is the relaxation parameter.
- Symmetric Successive Over-Relaxation (SSOR).